Section 4 solution

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4 Exercises

4.1 Envelope Theorem

A student wants to maximize her grade (g) on an essay. She selects her time spent writing (t) in order to achieve her top grade. Her time spent writing is a function of the noise at her local library (q), which is out of her control (or *exogenous*):

$$g = -(2t - q)^2 - t^2 + 100$$

In short, she is solving the following problem: $\max_t g(t; q)$. The FOC for this problem is given by: $-4(2t - q) - 2t = 0$ (check this). From our FOCs, we can solve for our optimal t in terms of q, finding that $t^*(q) = \frac{2}{5}q$. Using this information, calculate the derivative of g with respect to q at the optimal level of t (i.e., the rate at which her grade changes in response to a change in library "quality") in two different ways:

1. Plug in $t^*(q) = \frac{2}{5}q$ for t into our original expression for g, and then differentiate with respect to q.
2. (Envelope Theorem) Calculate the partial derivative $\frac{\partial g}{\partial q}$, and evaluate when $t = t^*(q)$.
3. How do these derivatives compare?

ANSWER:
1. In this case, $g(t^*(q), q) = -\left(\frac{4q}{5} - q\right)^2 - \left(\frac{2q}{5}\right)^2 = -\frac{q^2}{5}$. Taking the derivative, we get:

$$\frac{dg(t^*(q), q)}{dq} = -\frac{2}{5}q$$

2. Here, it's easy to see that $\frac{\partial g(t^*,q)}{\partial q} = 2(2t^* - q)$. Notice that if we substitute in $t^*(q) = \frac{2}{5}q$ for t^* above, we get $\frac{\partial g(t^*,q)}{\partial q} = 2 * (2 * \frac{2}{5}q - q) = -\frac{2}{5}q$.
3. They are the same! We find that increased noise of the library leads to a decrease in your grades.

4.2 Constrained Optimization

Each day, Elisabetta has quasi-linear utility over leisure l and consumption c. In particular, she maximizes $U(l, c; w) = \phi(l) + c + \psi(w)$. She can only spend her endowment M and wages earned wh, where w is the hourly wage rate and $h = 24 - l$ are her hours spent working. The price of consumption is p.

1. Derive Elisabetta's budget constraint.
2. Write down the maximization problem of the worker with respect to c and l with the relevant constraints. (Assume that the budget constraint is satisfied with equality.)

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3. Write down the Lagrangian, and derive the first order conditions with respect to c, l, and λ.
4. Simplify the problem to a system of two equations and two unknowns c^* and l^*.
5. Assume that $\phi'(l^*) > 0$ and $\phi''(l^*) < 0$ [Utility increases with more leisure but at a decreasing rate]. What would happen with the optimal leisure l^*if w increases? Provide intuition. If time permits, use IFT to formally find: $\frac{\partial l^*}{\partial w}$.

ANSWER:

1. Elisabetta's budget constraint:

$$pc + wl \leq M + 24w$$

2. The worker's problem:

$$\max_{c,l} \phi(l) + c + \psi(w)$$

$$s.t. \quad pc + wl = M + 24w$$

3. $$\mathcal{L}(c, l, \lambda; p, w, M) = \phi(l) + c + \psi(w) - \lambda(pc + wl - M - 24w)$$

Our FOCs:

$$\bigtriangledown\mathcal{L} = \begin{bmatrix} 1 - \lambda p \\ \phi'(l) - \lambda w \\ -pc - wl + M + 24w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

4. Solving for $\lambda^* = \frac{1}{p}$, we get:

$$\phi'(l^*) - \frac{w}{p} = 0$$

$$c^* = \frac{M + w(24 - l^*)}{p}$$

5. We know that:

$$\phi'(l^*) - \frac{w}{p} = 0 \qquad \Longrightarrow \qquad \phi'(l^*) = \frac{w}{p}$$

Thus, if w increases (ceteris paribus), l^*must decrease in order to increase $\phi'(l^*)$ and preserve the optimal (first order) condition.

Here the "f" from the IFT is vector valued function $\bigtriangledown\mathcal{L}$.

$$\bigtriangledown\mathcal{L} = \begin{bmatrix} 1 - \lambda p \\ \phi'(l) - \lambda w \\ -pc - wl + M + 24w \end{bmatrix}$$

Which is continuously differentiable for all positive c, l, and λ. Moreover, the Jacobian matrix of a gradient is the Hessian of the original scalar valued function ($\mathcal{L}$), for which we need to show that its determinant is non-zero.

$$H_L = \begin{bmatrix} \frac{\partial^2 L}{\partial c^2} & \frac{\partial^2 L}{\partial c \partial l} & \frac{\partial^2 L}{\partial c \partial \lambda} \\ \frac{\partial^2 L}{\partial l \partial c} & \frac{\partial^2}{\partial l^2} & \frac{\partial^2 L}{\partial l \partial \lambda} \\ \frac{\partial^2 L}{\partial \lambda \partial c} & \frac{\partial^2 L}{\partial \lambda \partial l} & \frac{\partial^2 L}{\partial \lambda^2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -p \\ 0 & \phi''(l) & -w \\ -p & -w & 0 \end{bmatrix}.$$

The determinant of H_L gives use the denumerator of the IFT. To find the numerator let's first write out the Hessian again, but for the second column (that column corresponds to the partial $\frac{\partial}{\partial l}$ taken on each term in that column), we instead take the partial with respect to w:

$$H_L^* = \begin{bmatrix} \frac{\partial^2 L}{\partial c^2} & \frac{\partial^2 L}{\partial l \partial w} & \frac{\partial^2 L}{\partial c \partial \lambda} \\ \frac{\partial^2 L}{\partial l \partial c} & \frac{\partial^2}{\partial l \partial w} & \frac{\partial^2 L}{\partial l \partial \lambda} \\ \frac{\partial^2 L}{\partial \lambda \partial c} & \frac{\partial^2 L}{\partial l \partial w} & \frac{\partial^2 L}{\partial \lambda^2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -p \\ 0 & -\lambda & -w \\ -p & (24-l) & 0 \end{bmatrix}.$$

Thus

$$\frac{\partial l^*}{\partial w} = -\frac{|H_L^*|}{|H_L|} = \frac{\lambda}{\phi''(l)} < 0$$

So, when w, the opportunity cost of leisure increases, the amount of leisure decreases.